

Scraftie

Personal Music Streaming Server

Initial Build Documentation

Historical engineering record of the initial server build, including implementation, failures, troubleshooting, verification, current architecture, and planned future work.

README.md

Scraftie — Personal Music Streaming Server

A private, self-hosted personal music streaming server built from an older HP laptop.

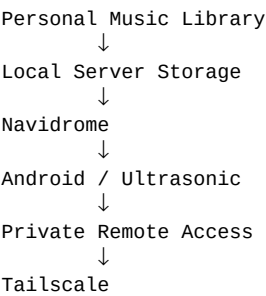
Current status: ☐ Functional initial build

Scraftie is a personal music server built incrementally from an older HP 15-da0037nr laptop.

The server runs Debian 13 (x86_64) with Navidrome providing the music-streaming service. Remote administration is performed from a Fedora workstation over SSH, while remote music access is provided through Tailscale.

The system is intended for private personal use rather than as a public streaming service.

Project Goal



The initial build focused on getting the complete pipeline working before attempting to make the server fully headless or building a custom client.

Current Status

Component	Status
Debian 13 server	☒ Completed
Music migration	☒ Completed
Navidrome	☒ Completed
SSH administration	☒ Completed
Tailscale networking	☒ Completed
Android streaming	☐ Verified
Cellular streaming through Tailscale	☐ Verified
Lid-close server operation	☐ Configured and tested

Component	Status
GNOME suspend prevention	☐ Configured
Fully headless server	☐ Planned
Custom Android client	☐ Planned
Last.fm integration	☐ Planned

Features

Currently implemented

- Personal music library hosted locally
- Navidrome music streaming
- Debian 13 server
- SSH administration
- Private Tailscale networking
- Android streaming through Ultrasonic
- Remote playback over cellular data
- Laptop lid configured so closing the lid does not intentionally suspend the server
- Samsung T7 Shield retained as the master music copy

Not currently implemented

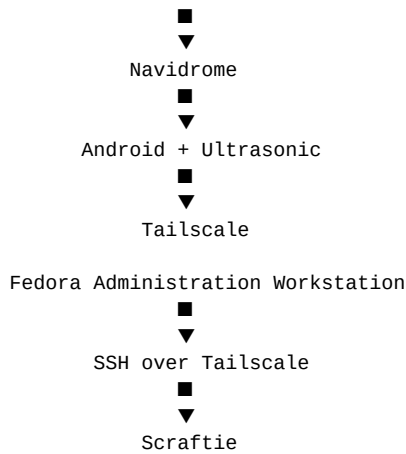
- Public internet exposure
- Tailscale Funnel
- Public port forwarding
- Custom Android application
- Automated recommendation system
- Last.fm integration
- Automated upload/import workflow
- Automated offline music rotation
- Fully headless operating-system configuration

Architecture

```

Samsung T7 Shield (Master Music Copy)
  ■
  ▼
Scraftie /music/06_SD_Card

```



Hardware

Model: HP 15-da0037nr

The exact CPU, RAM, storage-device model/capacity, networking hardware, and original operating system were not established in the retained build record and are intentionally not guessed here.

The Samsung T7 Shield is used as the master/original music copy.

Software Stack

Component	Role
Debian 13	Server operating system
Navidrome	Music streaming server
OpenSSH	Remote administration
Tailscale	Private remote networking
Ultrasonic	Android Navidrome/Subsonic-compatible client
Fedora	Administration workstation
Android / Motorola Edge+ 2023	Remote streaming client

Music Storage

The original music directory was renamed from:

OG SD Card

to:

OG_SD_Card

The server-side library is:

```
/music/OG_SD_Card
```

Verification:

```
find /music/OG_SD_Card -type f | wc -l  
du -sh /music/OG_SD_Card
```

Results:

```
4,669 files  
26G
```

The Samsung T7 Shield retains the original/master copy.

Navidrome

Configuration:

```
DataFolder = "/var/lib/navidrome"  
MusicFolder = "/music/OG_SD_Card"
```

The initial service failure was:

```
Error disabling foreign_keys  
unable to open database file: no such file or directory  
Database could not be opened!
```

The fix was:

```
sudo mkdir -p /var/lib/navidrome  
sudo chown -R navidrome:navidrome /var/lib/navidrome  
sudo chown -R navidrome:navidrome /music/OG_SD_Card  
sudo systemctl restart navidrome
```

Navidrome subsequently became active and scanned the library.

Remote Access

Tailscale connects Scaftie, Fedora, and the Android phone through a private network.

The real Tailscale address is intentionally omitted:

```
ssh cprquecian@<SCARFTIE_TAILSCALE_IP>
```

Tailscale Funnel is not part of the intended architecture.

Android Verification

The Motorola Edge+ 2023 used Ultrasonic to connect to Navidrome.

The decisive remote test was performed with Wi-Fi disabled. Music continued playing over cellular data through Tailscale.

This demonstrated actual remote connectivity rather than merely local-Wi-Fi playback.

Lid and Power Management

The server initially stopped responding over Tailscale when the laptop lid was closed.

The intended `/etc/systemd/logind.conf` settings became:

```
HandleLidSwitch=ignore
HandleLidSwitchExternalPower=ignore
HandleLidSwitchDocked=ignore
LidSwitchIgnoreInhibited=yes
```

GNOME did not contain the expected `lid-close-ac-action` key.

Available keys were inspected with:

```
gsettings list-keys org.gnome.settings-daemon.plugins.power
```

The AC inactivity action was changed with:

```
gsettings set org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type 'nothing'
```

Verification:

```
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
```

Result:

```
'nothing'
```

Security Model

- Personal/private use
- SSH administration
- Tailscale private networking
- No intentional public port forwarding

- No Tailscale Funnel
- Credentials excluded from GitHub
- No secrets committed to the repository

Documentation

- docs/installation.md — chronological initial build
- docs/architecture.md — current architecture and service model
- docs/troubleshooting.md — failures, diagnosis, fixes, and verification
- docs/roadmap.md — future work

Lessons Learned

1. A successfully installed application can still fail as a systemd service.
2. Service accounts need access to their data and application directories.
3. Laptop hardware introduces server-specific power-management problems.
4. Remote access should be tested from outside the local network.
5. Configuration should be inspected on the actual installed system instead of assumed.
6. Troubleshooting history is valuable engineering documentation.

Current Limitations

- GNOME remains installed.
- The server is not yet fully headless.
- The custom Android client does not yet exist.
- Recommendation and listening-analysis features are not implemented.
- Last.fm integration is not implemented.
- Automated music import workflows are not implemented.
- Public-facing streaming is not intended.

installation.md

Scraftie Installation and Initial Build Record

This document records the initial construction of Scraftie from an older HP laptop into a functioning private music streaming server.

This is a historical build record. It intentionally records mistakes and troubleshooting rather than presenting a clean-room installation procedure.

1. Server Hardware

Model: HP 15-da0037nr

The exact CPU, RAM, storage-device model/capacity, networking hardware, and original operating system were not established in the retained build record and are therefore not reproduced here.

2. Operating System

Scraftie runs:

```
Debian 13  
x86_64
```

The running kernel was verified as:

```
Linux Scraftie 6.12.107+deb13-amd64
```

A desktop/GNOME environment was accidentally installed during the initial setup. GNOME was temporarily retained because it was useful during setup and troubleshooting.

The long-term plan is to remove the desktop environment and operate Scraftie as a fully headless server.

The exact Debian installer ISO filename and checksum were not retained in the available build record and are therefore not documented.

3. Debian User and Sudo

The primary administration user was:

```
cprquecian
```

An initial problem occurred where the user did not have expected sudo access.

A root shell was used to correct the account.

The user was added to Debian's sudo group:

```
usermod -aG sudo cprquecian
```

The user then logged out/reconnected or rebooted so the new group membership applied.

Sudo access subsequently worked.

4. Music Library Migration

The original music directory:

```
OG SD Card
```

was renamed to:

```
OG_SD_Card
```

The server music directory was created:

```
sudo mkdir -p /music
```

The final server library path is:

```
/music/OG_SD_Card
```

Verification:

```
find /music/OG_SD_Card -type f | wc -l
```

Result:

```
4669
```

Size:

```
du -sh /music/OG_SD_Card
```

Result:

```
26G
```

5. Master Copy

The Samsung T7 Shield retains the original/master music copy.

The server maintains its own copy under:

```
/music/OG_SD_Card
```

The T7 is intentionally retained as the master so the server is not the only copy.

6. Navidrome Installation

The first installation/download attempt returned:

```
HTTP 404
```

The Navidrome GitHub release API was investigated.

Architecture was determined with:

```
uname -m
```

The required architecture was:

```
amd64
```

The appropriate Debian amd64 Navidrome package was then identified and installed.

7. Navidrome Directories and Configuration

Navidrome uses:

```
/opt/navidrome  
/var/lib/navidrome  
/etc/navidrome/navidrome.toml
```

Configuration:

```
DataFolder = "/var/lib/navidrome"  
MusicFolder = "/music/OG_SD_Card"
```

A dedicated system account named navidrome was created.

8. First Navidrome Failure

Initial startup failed with:

```
Error disabling foreign_keys
```

```
unable to open database file: no such file or directory
Database could not be opened!
```

The missing/unavailable Navidrome data directory was the cause identified during the build.

9. Navidrome Fix

```
sudo mkdir -p /var/lib/navidrome
sudo chown -R navidrome:navidrome /var/lib/navidrome
sudo chown -R navidrome:navidrome /music/OG_SD_Card
sudo systemctl restart navidrome
```

Navidrome subsequently became active and scanned the library.

The web interface opened successfully and an administrator account was created. The password is intentionally omitted.

10. SSH

OpenSSH was installed and the SSH service was enabled and tested.

Administration is performed from Fedora.

Connection format:

```
ssh cprquecian@<SCARFTIE_TAILSCALE_IP>
```

The real Tailscale address is omitted.

A remote shell can be closed with:

```
exit
```

or:

```
ctrl+D
```

A remote reboot terminates the SSH connection while the server restarts. SSH can be re-established after boot.

11. Tailscale

Tailscale was configured on:

- Scraftie

- Fedora
- Motorola Edge+ 2023

The daemon was enabled and Tailscale was brought online:

```
tailscale up
```

Connectivity was tested:

```
tailscale ping <SCARFTIE_TAILSCALE_IP>
```

The result was a successful pong.

SSH over Tailscale was verified.

12. Android Streaming

The Motorola Edge+ 2023 used Ultrasonic.

Navidrome playback was successfully verified.

Wi-Fi was then disabled and music continued playing over cellular data through Tailscale.

This proved the server was remotely reachable rather than merely functioning on local Wi-Fi.

13. Lid Behavior

Closing the laptop lid initially caused the server to stop responding over Tailscale.

/etc/systemd/logind.conf initially contained commented settings:

```
#HandleLidSwitch=ignore
#HandleLidSwitchExternalPower=ignore
#HandleLidSwitchDocked=ignore
```

The intended settings became:

```
HandleLidSwitch=ignore
HandleLidSwitchExternalPower=ignore
HandleLidSwitchDocked=ignore
LidSwitchIgnoreInhibited=yes
```

14. GNOME Power Management

The expected GNOME key:

```
lid-close-ac-action
```

did not exist.

The error was:

```
No such key "lid-close-ac-action"
```

Available power-management keys were inspected:

```
gsettings list-keys org.gnome.settings-daemon.plugins.power
```

Relevant keys included:

```
sleep-inactive-ac-timeout
sleep-inactive-ac-type
sleep-inactive-battery-timeout
sleep-inactive-battery-type
```

The initial AC inactivity configuration was:

```
sleep-inactive-ac-type = 'suspend'
sleep-inactive-ac-timeout = 900
```

The final change was:

```
gsettings set org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type 'nothing'
```

Verification:

```
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
```

Result:

```
'nothing'
```

The gsettings change takes effect immediately and does not require a reboot.

15. Current State

At the end of the initial build:

- Debian 13 is running.
- Scraftie is reachable through Tailscale.
- SSH administration works.
- Navidrome is active.

- The music library is available to Navidrome.
- Android/Ultrasonic playback works.
- Cellular playback through Tailscale was verified.
- Lid-close/power behavior was configured.
- GNOME remains installed.
- A fully headless conversion remains future work.

troubleshooting.md

Scraftie Troubleshooting Record

This document preserves the actual problems encountered during the initial build.

1. Navidrome Installation Returned HTTP 404

Problem

The first Navidrome installation/download attempt returned:

```
HTTP 404
```

Investigation

The Navidrome GitHub release API was investigated instead of assuming the URL was correct.

The architecture was checked:

```
uname -m
```

Result:

```
amd64
```

The appropriate Debian amd64 release package was identified and installed.

Result

Navidrome installed successfully.

2. Navidrome Could Not Open Its Database

Problem

Navidrome failed with:

```
Error disabling foreign_keys
unable to open database file: no such file or directory
Database could not be opened!
```

Diagnosis

The configured data directory:

```
/var/lib/navidrome
```

was not available correctly for the service.

Fix

```
sudo mkdir -p /var/lib/navidrome
sudo chown -R navidrome:navidrome /var/lib/navidrome
sudo chown -R navidrome:navidrome /music/OG_SD_Card
sudo systemctl restart navidrome
```

Verification

Navidrome became active and scanned the library.

The web interface opened and an administrator account was created.

3. Sudo Access

Problem

cprquecian initially did not have expected sudo access.

Fix

A root shell was used to add the user to the sudo group:

```
usermod -aG sudo cprquecian
```

The user then logged out/reconnected or rebooted so group membership applied.

Sudo access subsequently worked.

4. Lid Closure Caused Server Availability Problems

Problem

Closing the Scraftie laptop lid caused the machine to stop responding over Tailscale.

Investigation

/etc/systemd/logind.conf contained:

```
#HandleLidSwitch=ignore
#HandleLidSwitchExternalPower=ignore
#HandleLidSwitchDocked=ignore
```

Configuration

The intended configuration became:

```
HandleLidSwitch=ignore
HandleLidSwitchExternalPower=ignore
HandleLidSwitchDocked=ignore
LidSwitchIgnoreInhibited=yes
```

5. Expected GNOME Lid Setting Did Not Exist

An expected key:

```
lid-close-ac-action
```

returned:

```
No such key "lid-close-ac-action"
```

The available GNOME power keys were inspected:

```
gsettings list-keys org.gnome.settings-daemon.plugins.power
```

Relevant keys:

```
sleep-inactive-ac-timeout
sleep-inactive-ac-type
sleep-inactive-battery-timeout
sleep-inactive-battery-type
```

Initial AC configuration:

```
sleep-inactive-ac-type = 'suspend'
sleep-inactive-ac-timeout = 900
```

Final configuration:

```
gsettings set org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type 'nothing'
```

Verification:

```
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
```

Result:

```
'nothing'
```

6. Temporary GNOME Display Problem

During the power/lid troubleshooting, the physical display eventually showed a blinking underscore.

SSH continued working.

This established that the underlying server was still operational even though the local graphical display was not behaving normally.

7. Display Diagnostics

The following diagnostics were performed:

```
systemctl status gdm3 --no-pager
systemctl --user status gnome-session.target --no-pager
loginctl list-sessions
loginctl show-session 4 -p Type -p Display -p Remote -p State -p Desktop -p IdleHint
pgrep -a gnome-shell
```

Findings:

- GDM was active.
- GNOME session was active.
- GNOME Shell was running.
- The local session was Wayland.
- SSH remained operational.
- DRM-related messages appeared in the graphical session.

8. DRM Messages

Relevant messages included:

```
Cursor update failed: drmModeAtomicCommit: Invalid argument
```

and:

```
Failed to disable device '/dev/dri/card0': drmModeAtomicCommit: Permission denied
```

These messages are preserved as observed diagnostic output.

The build record does not establish either message as the definitive root cause of the display issue.

9. Recovery

A clean reboot restored the normal GNOME login screen and desktop.

The important conclusion was:

The server itself remained operational; the temporary issue affected the graphical display/session.

10. Remote Connectivity

Tailscale was tested with:

```
tailscale ping <SCARFTIE_TAILSCALE_IP>
```

and returned a successful pong.

SSH from Fedora to Scraftie was also verified.

11. Cellular Streaming

Ultrasonic successfully streamed music from Navidrome.

With Wi-Fi disabled on the Motorola Edge+ 2023, playback continued over cellular through Tailscale.

This verified the end-to-end remote path.

12. Useful Service Diagnostics

```
systemctl status navidrome
systemctl status tailscaled
systemctl status ssh
```

```
systemctl is-active navidrome tailscaled ssh
```

```
systemctl is-enabled navidrome
systemctl is-enabled tailscaled
systemctl is-enabled ssh
```

```
systemctl --failed
```

```
journalctl -u navidrome -n 50
journalctl -u navidrome -f
```

The first group checks service state; `is-active` provides a quick status check; `is-enabled` checks boot enablement; `systemctl --failed` identifies failed units; and `journalctl` provides Navidrome logs.

Troubleshooting Pattern

The initial build repeatedly followed:

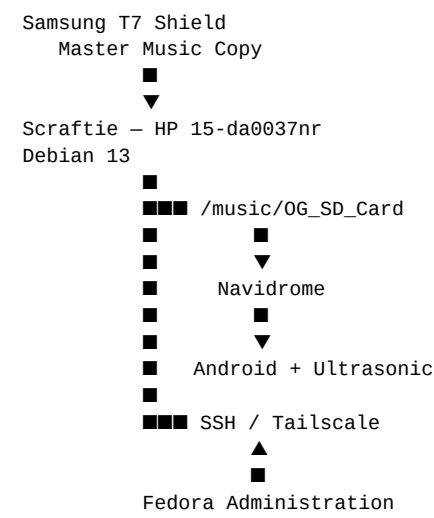
```
Problem
↓
Inspect actual system state
↓
Identify cause
↓
Apply targeted fix
↓
Restart/retest when required
↓
Verify from the user's actual workflow
```

This process is intentionally preserved as part of the project record.

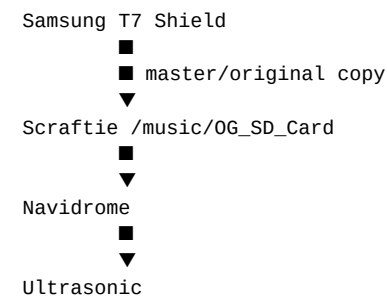
architecture.md

Scraftie Architecture

Current Architecture

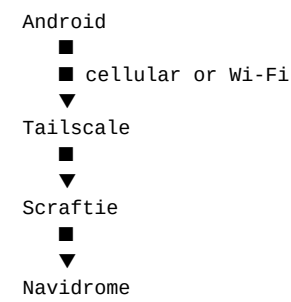


Current Data Flow

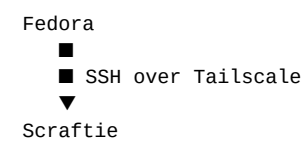


The T7 remains the master copy. The server has its own copy.

Network Flow

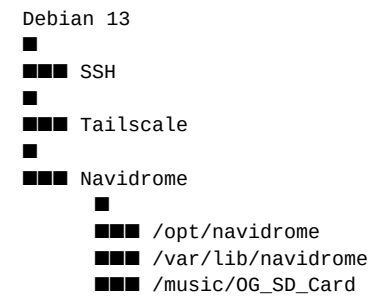


Administration:



Tailscale Funnel is intentionally not part of the architecture.

Service Layout



Current Configuration

Item	Current state
Host	HP 15-da0037nr
OS	Debian 13
Architecture	x86_64
Kernel	6.12.107+deb13-amd64
Administration user	cprquecian
Music path	/music/OG_SD_Card
Navidrome data	/var/lib/navidrome
Navidrome application	/opt/navidrome
Navidrome configuration	/etc/navidrome/navidrome.toml
Remote administration	SSH
Private networking	Tailscale
Android client	Ultrasonic
Master music copy	Samsung T7 Shield
Desktop environment	GNOME
Fully headless	No

Service Commands

```
systemctl status navidrome  
systemctl status tailscaled  
systemctl status ssh
```

```
systemctl is-active navidrome tailscaled ssh
```

```
systemctl is-enabled navidrome  
systemctl is-enabled tailscaled  
systemctl is-enabled ssh
```

```
systemctl --failed
```

```
journalctl -u navidrome -n 50  
journalctl -u navidrome -f
```

The actual Tailscale IP addresses and tailnet name are intentionally omitted from public documentation.

roadmap.md

Scraftie Roadmap

This document contains future work discussed during the initial build. Planned features are deliberately not presented as completed.

Server

Status	Item
☐	Remove GNOME
☐	Fully headless operation
☐	SMART monitoring
☐	Additional maintenance automation

Remote Access

Status	Item
☐	Evaluate Tailscale Serve
☐	Evaluate reverse-proxy architecture
☐	Evaluate HTTPS

These are future considerations, not current configuration.

Custom Android Application

The long-term goal is a custom Android client designed around the user's own library.

Tap-to-Play

Tap song
↓
Immediately play

Album Queue

Selected track
↓
Play selected track
↓
Queue remaining tracks from album

End of Album

Possible future behavior:

Album finishes
■■■ Loop album
■■■ Continue with similar music

Multi-Select

Allow multiple tracks to be selected and played directly.

Background Playback

Potential features:

- Background playback
- Lock-screen controls
- Improved queue behavior

Listening Analysis

Potential data:

- Play count
- Last played
- Skip rate
- Time of day
- Month/season
- Favorites
- Album context
- Artist context
- Album sessions

Potential analysis windows:

- 24 hours
- 30 days
- 3 months
- 1 year

- 2 years
- 3+ years

Music Lifecycle

Potential future categories:

Age since active listening	Possible category
0-12 months	Active
1-2 years	Forgotten favorites / archive
2-3 years	Aging archive
3+ years	Possible external archive / rotation

These are application concepts, not implemented functionality.

Last.fm Integration

Planned relationship:

```

Last.fm
  ↓
Similarity / recommendation metadata
  ↓
Scraftie application
  ↓
Check local library
  ↓
Only recommend locally available tracks

```

The application should eventually filter recommendations so it only offers music that actually exists in the user's library.

Last.fm integration is not currently implemented.

Future Music Import Workflow

A possible future workflow:

```

Incoming Music
  ↓
Personal Upload API
  ↓
Incoming / staging area
  ↓
Library processing
  ↓
Navidrome library
  ↓
Automated scan

```

Planned concepts include:

- Personal upload API
- Incoming → Library workflow
- Automated Navidrome scanning
- Automated offline music rotation

Repository Structure

```
Scraftie-Music-Server/  
■■■ README.md  
■■■ docs/  
■   ■■■ installation.md  
■   ■■■ architecture.md  
■   ■■■ troubleshooting.md  
■   ■■■ roadmap.md  
■■■ scripts/  
■■■ config/
```

Only actual project files and non-secret configuration should be committed.

Status Summary

☐ Completed

- Debian 13
- Music migration
- Navidrome
- SSH
- Tailscale
- Android/Ultrasonic playback
- Cellular streaming test
- Lid-close configuration
- GNOME suspend prevention
- Remote administration

☐ In Progress

The initial infrastructure build is functional. Further work is primarily refinement and future application development.

☐ **Planned**

- Headless conversion
- SMART monitoring
- HTTPS
- Tailscale Serve/reverse-proxy evaluation
- Custom Android client
- Tap-to-play
- Album-aware queues
- Multi-select
- Background playback
- Lock-screen controls
- Listening history
- Listening analysis
- Last.fm integration
- Library-aware recommendations
- Automated offline music rotation
- Personal upload API
- Incoming → Library workflow
- Automated Navidrome scanning